

general population," says professor Ranjan, confident of conquering all the previous limitations.

A complete solution. Taking advantage of MEMS technology-based accelerometer, CePal senses movements with high accuracy. "We kept in mind that cerebral palsy patients do not have very fine control over movements. We put logic in the firmware to detect the tilts," explains professor Ranjan. Considering each patient's movement capability and behavioural pattern may differ, CePal is also able to tune parameters to match needs. "CePal provided flexibility in firmware to adjust to the needs, making it easy to be customisable at the user's end," says professor Ranjan.

Also, a work in progress would map the user-generated sound to meaningful voice outputs. A microphone is attached to the device that sends the sound input to the central processing unit via a wireless link, which is then mapped to

prerecorded voice and corresponding meaningful response to be fed as the output to the speaker.

Cost-effective. CePal is much cheaper than other assistive technology products in the market. Use of the latest hardware components, including MEMS-based sensors, has helped to minimise the number of components, reducing the overall costs. The use of Open Source technology has also made CePal cost-effective. "We used Open Source embedded system development tools and all development is done on Linux platform. In particular, we used GNU GCC cross compilers for firmware development, which helped to cut costs," says professor Ranjan.

Easy to use. Usability of the device is of paramount importance for cerebral palsy patients and CePal ensures ease of use with its light weight. "We have used recent developments in technology to keep the weight low. The use of a microcontroller helped to

reduce the component count. Further, the complete architecture is designed on a light-weight PCB to facilitate the user in using it without any hassle," says professor Ranjan.

Road ahead

National Trust will test CePal on a large scale in various parts of India. It has already won the *HP Innovate Award* in 2009. Professor Ranjan and National Trust hope to take CePal to the next level, making it the guiding light for aged people.

"We are looking at making CePal of great assistance to old people facing difficulty in the movement of limbs and people suffering from arthritis, paralysis and other physical impairments like carpal tunnel syndrome caused by excessive use of keyboard and mouse," says professor Ranjan. ●

The author is a business correspondent at EFY Bengaluru

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